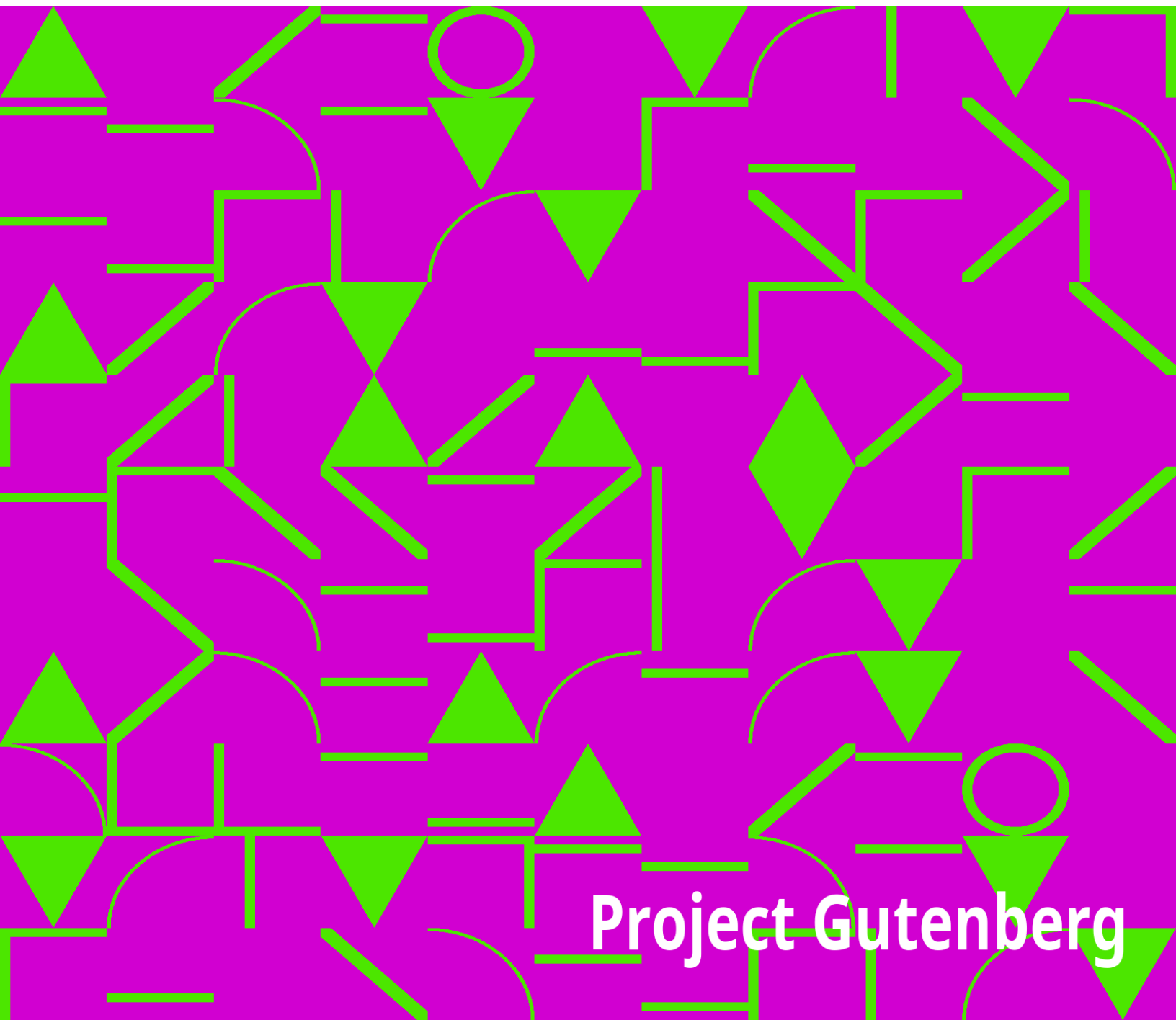


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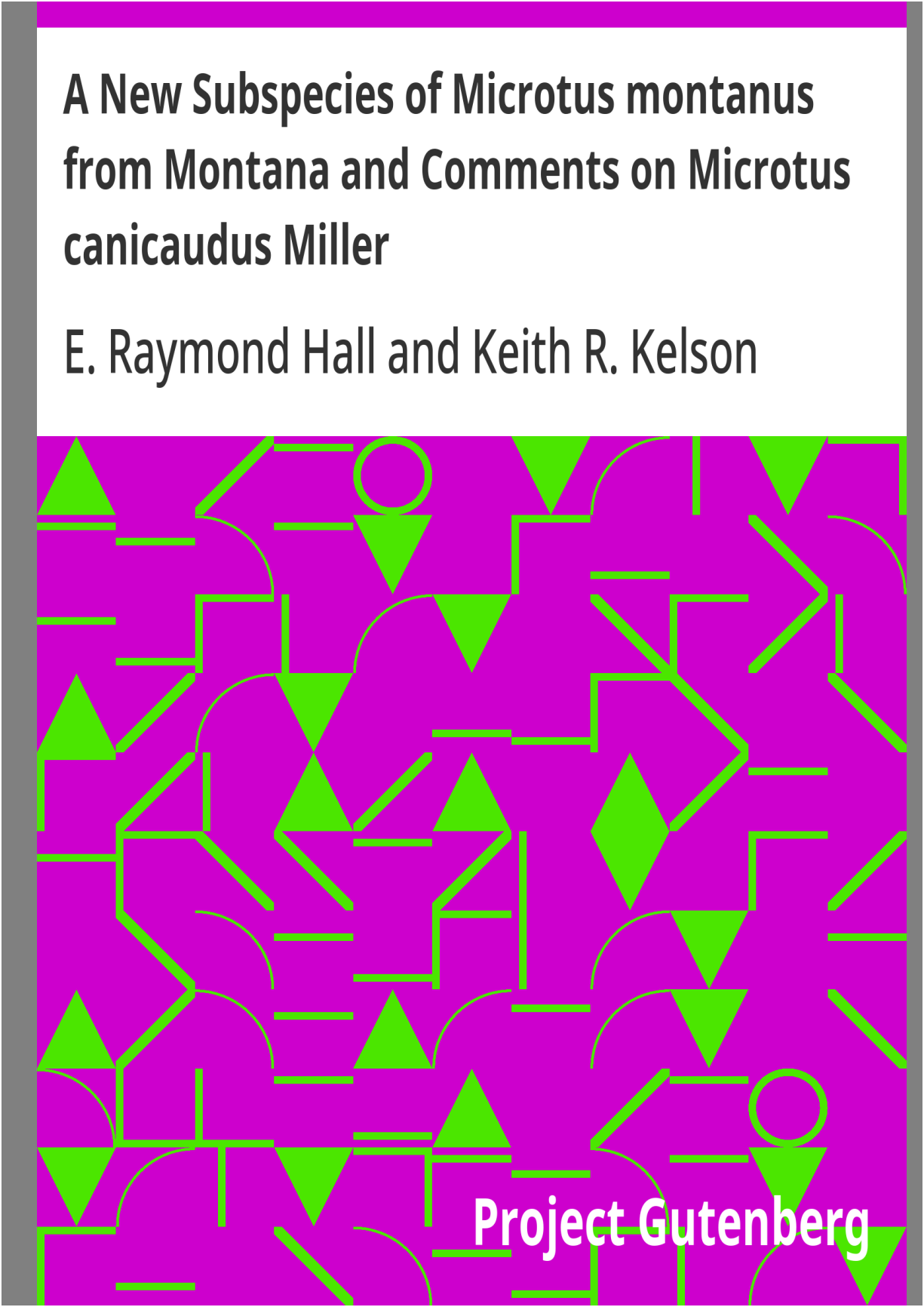
E. Raymond Hall and Keith R. Kelson



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*** START OF THE PROJECT GUTENBERG EBOOK A NEW
SUBSPECIES OF *MICROTUS MONTANUS* FROM MONTANA AND
COMMENTS ON *MICROTUS CANICAUDUS* MILLER ***

A New Subspecies of *Microtus* *montanus* from Montana and Comments on *Microtus* *canicaudus* Miller

BY

E. RAYMOND HALL and KEITH R. KELSON

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A New Subspecies of *Microtus montanus* from Montana and Comments on *Microtus canicaudus* Miller

BY

E. RAYMOND HALL AND KEITH R. KELSON

IN 1949, for the Museum of Natural History of the University of Kansas, Mr. John A. White collected two specimens of the species *Microtus montanus* in the Bitterroot Valley of Montana, that did not fit the description of any named subspecies. These were laid aside until we could examine the additional specimens from Montana in the Biological Surveys collection in the United States National Museum, some of which previously had been reported by Bailey (N. Amer. Fauna, 17:31, June 6, 1900) under the name *Microtus nanus canescens* Bailey [= *Microtus montanus canescens*]. Our examination reveals that the animals from the Bitterroot and Flathead valleys belong to an heretofore unrecognized subspecies which is named and described below.

***Microtus montanus pratincolus* new subspecies**

Type.—Female, adult, skull and skin, No. 34004, Univ. Kansas, Mus. Nat. Hist.; from 6 mi. E Hamilton, 3700 ft., Ravalli County, Montana; obtained on August 14, 1949, by John A. White; original number 477.

Geographic distribution.—Flathead and Bitterroot valleys of western Montana.

Diagnosis.—Size small for the species (see measurements). Color: Essentially as in *Microtus montanus nanus*. Skull: Small, slender, and comparatively smooth; rostrum moderately depressed distally; nasals moderately inflated distally and extending posteriorly not quite to tips of premaxillary tongues; nasals usually truncate posteriorly, but rounded in some individuals; premaxillary tongues terminating posteriorly in a short medial spine; zygomatic arches lightly constructed and usually more widely spreading posteriorly than anteriorly; interparietal comparatively long and terminating

in a small, but distinct, medial spine, otherwise approximately rectangular in shape; exposed parts of upper incisors short and, for the species, only slightly procumbent; molar dentition weak and, in most specimens, especially so posteriorly; tympanic bullae large and well inflated, especially ventrolaterally; basioccipital narrow owing to the encroachment of the tympanic bullae.

Comparison.—Among named forms, *Microtus montanus pratincolus* most closely resembles *Microtus montanus nanus*. The geographic range of *M. m. nanus* adjoins that of *M. m. pratincolus* on three sides; there is no conspecific subspecies adjoining the range of *M. m. pratincolus* on the north. From *M. m. nanus*, *M. m. pratincolus* differs as follows (measurements are all of males, those of *M. m. nanus* being of nine topotypes and near topotypes from central Idaho): size smaller (149 mm. as opposed to 165), tail shorter (37 as opposed to 39), hind foot shorter (19 as opposed to 20), upper molar series shorter (expressed as a percentage of basilar length, 25.5 as opposed to 26.3), mastoidal region broader (expressed as a percentage of basilar length, 48.6 as opposed to 46.7), braincase slightly more vaulted (depth of braincase expressed as a percentage of basilar length, 31.3 as opposed to 30.4) and more inflated laterally; tympanic bullae more inflated, this inflation being the most conspicuous difference between the two subspecies. The tympanic bullae of *M. m. pratincolus* have approximately a fourth more volume than those of *M. m. nanus*.

Remarks.—Northwardly in the Bitterroot Valley, specimens of *M. m. pratincolus* morphologically approach *M. m. nanus*, especially in the reduced degree of inflation of the tympanic portion of the bullae. On geographic grounds we think that the geographic range of *M. m. pratincolus* extends southward to the southern end of the Bitterroot Valley; we have not seen specimens from that area. Although we have not examined the specimen reported upon by Davis (Murrelet, 18:26, September 4, 1937) from Canyon Creek, "a few miles west of Hamilton", Montana, we think that it will be found to belong to *M. m. pratincolus*.

Our examination of specimens from localities in Montana east of the range here ascribed to *M. m. pratincolus* indicates that, among named kinds of *Microtus*, those specimens are best referred to *M. m. nanus*. These specimens are listed below under comparative materials. It should be mentioned here that although Bailey (*loc. cit.*) applied the name *Microtus nanus canescens* to Montanan specimens from Flathead Lake and Hot Springs Creek, the subspecies *M. montanus canescens* now is thought to be restricted to Washington and the adjoining part of British Columbia; *M. m. canescens* does not occur so far east as Montana.

Grateful acknowledgment is made to those persons in charge of the Biological Surveys collection for permission to study the specimens in that collection, and to the Kansas Endowment Association for support of the field work which yielded the specimens from six miles east of Hamilton, Montana. The study here reported upon was aided also by a contract between the Office of Naval Research, Department of the Navy, and the University of Kansas (NR 161–791).

Measurements.—The following measurements in millimeters are those of the holotype and the average, maximum, and minimum, respectively, of eleven adult males from various places in the range of the subspecies. Except as noted below, we are unable to detect significant morphological differences in the populations sampled and believe that pooling of the measurements is justifiable in this case. Measurements are: Total length, 129, 149 (156–141); length of tail-vertebrae, 27, 37 (39–31); length of hind foot, 18, 19 (20–18) (all preceding measurements are those of the collectors); basilar length, 22.2, 24.5 (25.7–23.4); greatest length of nasals, 6.7, 6.9 (7.4–6.4); zygomatic breadth, 14.2, 14.6 (14.9–13.9); mastoidal breadth, 11.3, 11.8 (12.3–10.8); alveolar length of upper molar series, 6.0, 6.2 (6.5–5.9); depth of braincase (shortest distance from ventral surface of basioccipito-basisphenoidal suture to the dorsal surface of the cranium, and not perpendicular to the long axis of the skull), 7.7, 7.7 (7.9–7.5); width of rostrum, 4.7, 4.8 (5.0–4.6); interorbital breadth, 3.2, 3.4 (3.6–3.2). Measurements of females, other than those of the holotype, are not given owing to the lack of sufficient material. Females, however, do not appear to differ appreciably in measurements from males.

Specimens examined (in U.S. Nat. Mus., Biol. Surv. Coll., except as otherwise indicated).—Total, 26, all from Montana, as follows: *Sanders Co.*: Hot-springs Cr., 4. *Lake Co.*: end of W arm Flathead Lake, 5; Ravalli, 8. *Ravalli Co.*: Florence, 2; 8 mi. NE Stevensville, 4000 ft., 1; Corvallis, 4; 6 mi. E Hamilton, 3700 ft., 2(K.U.).

Comparative materials (in U.S. Nat. Mus., Biol. Surv. Coll.).—*Microtus montanus nanus*: Total, 72, as follows: IDAHO: *Lewis Co.*: Nez Perce, 3. *Idaho Co.*: Seven Devils Mts., 3550 ft., 3. *Custer Co.*: Challis, 7; Pahsimeroi Mts. (9300 ft., 8; 9700 ft., 4), 12. *Butte Co.*: Lost River Mts., 1. *Canyon Co.*: Nampa, 1; Bowmont, 1. *Ada Co.*: Boise, 1. *Blaine Co.*: Sawtooth Lake, 2; Alturas Lake, 3. *Owyhee Co.*: Three Cr., 3. *Minidoka Co.*: Heyburn, 2. *Bannock Co.*: Pocatello, 4. *Bear Lake Co.*: Montpelier Cr., 3. MONTANA: *Fergus Co.*: Big Snowy Mts., 11. *Gallatin Co.*: West Fork of West Fork, Gallatin River, 1. *Park Co.*: Lamar River, 7000 ft., 2; Gardiner, 1. *Sweetgrass Co.*: "near" head Big Timber Cr., Crazy Mts., 1; Big Timber Cr., 5200 ft., Crazy Mts., 1; 14 mi. S Big Timber, 1; West Boulder Cr., 18 mi. SE Livingston, 2; McLeod, 1. *Carbon Co.*: Beartooth Mts., 2; Beartooth Lake, 1. WYOMING: *Park Co.*: N end Lake, Yellowstone Nat'l Park, 2.

***Microtus montanus canicaudus* Miller**

1897. *Microtus canicaudus* Miller, Proc. Biol. Soc. Washington, 11:67, April 21, type from McCoy, Willamette Valley, Polk County, Oregon.

In 1938 when one of us (Hall, Proc. Biol. Soc. Washington, 51:131–134, August 23, 1938) arranged several nominal species of *Microtus* as subspecies of the species *Microtus montanus*, *Microtus canicaudus* was not included because that writer had not examined representative specimens. In the U.S. Biological Surveys collection in the U.S. National Museum we have examined specimens of *M. m. canicaudus*, all from Oregon, as follows: Hood River (Catalogue Nos. 262583–262586); Canby (262577, 262578); Wapinitia (79985–79988); Sheridan (69779, 69780); McCoy (75834–75842, 77744); Salem (246736); Albany (161554); and Corvallis (242552). The four specimens from Wapinitia seem to be those that Bailey (N. Amer. Fauna, 17:29, June 6, 1900) listed as *Microtus montanus*. The diagnostic characters mentioned by Miller in the original description (Proc. Biol. Soc. Washington, 11:67, April 21, 1897) included the following: size approximately the same as in *Microtus [montanus] nanus*; upper parts yellowish; tail usually nearly uniform grayish above and below; auditory bullae much inflated; lateral pits at posterior edge of bony palate unusually shallow. Because the tails of the original series were understuffed and variously rotated, they seemed to be less sharply bicolored than is the case, as shown by subsequently collected specimens. Otherwise we find that the characters mentioned above differentiate *canicaudus* from its nearest relatives, *Microtus montanus canescens* to the northward, *M. m. nanus* to the eastward, and *M. m. montanus* to the southward. In *canicaudus* we have noted one additional differential character; the interpterygoid space is acuminate anteriorly. In this feature and in each of the other features mentioned above, intergradation with *Microtus montanus nanus* is seen in the specimens from Hood River and Wapinitia. In the specimens from Hood River the auditory bullae are only slightly less inflated than in those topotypes of *canicaudus* having the smallest bullae; there is appreciable variation in size of the bullae in the topotypes. Even so, the minimum size of bullae among the topotypes is larger than the maximum size in the specimens from Wapinitia. The four specimens from Wapinitia have the yellowish color of *canicaudus* to a considerable degree, and show intergradation between *canicaudus* and *nanus* in depth of the palatal pits

and shape of interpterygoid space. The slightly larger size of these specimens from Wapinitia suggests intergradation with *M. m. montanus*. The tympanic bullae in the specimens from Wapinitia seem to be smaller than in specimens of *canicaudus*, *nanus*, or *montanus*.

Because of the intergradation described above between *Microtus montanus nanus* and *M. canicaudus*, the latter should stand as *Microtus montanus canicaudus*.

Bailey (N. Amer. Fauna, 55:206, August 29, 1936) recorded *canicaudus* from Warm Springs in the Deschutes Valley of Oregon and from the state of Washington. Other authors also have recorded *canicaudus* from the state of Washington. Our examination of specimens leads us to conclude, as did Dalquest (Univ. Kansas Publ., Mus. Nat. Hist., 2:348, 349, April 9, 1948), that *canicaudus* does not occur in Washington. The reported occurrence of *M. canicaudus* at Warm Springs, Deschutes Valley, Oregon, seems to be the result of an error in identification. The specimens concerned seem to be two *Microtus longicaudus mordax* (Nos. 207060 and 207082 U.S.N.M.). They are labeled as collected at "Warm Springs (Mill Cr.—20 Mi. W of)". Bailey's (*op. cit.*, fig. 46, p. 209) map showing the distribution in Oregon of *Microtus mordax mordax* [= *Microtus longicaudus mordax*] has a locality-dot at Warm Springs itself. Bailey seems to have erred; he should have placed this dot 20 miles farther west, we think. When preparing his map (*op. cit.*, fig. 43, p. 205) showing the geographic distribution of *Microtus canicaudus*, Bailey seems to have misidentified these same two specimens as *M. canicaudus*, and for them placed a locality-dot on his map 20 miles east (instead of west) of Warm Springs. In brief, Bailey probably did not see any specimens of *canicaudus* or specimens of any other subspecies of *Microtus montanus* from Warm Springs.

Transmitted February 15, 1951.

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